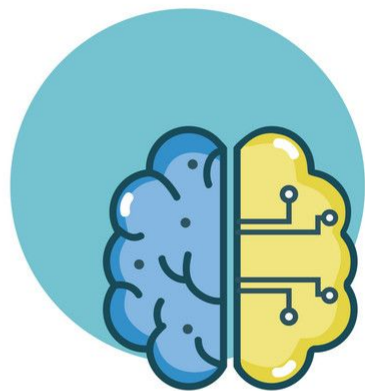


INTRODUCTION TO MACHINE LEARNING

INTRODUCTION



Elisa Ricci



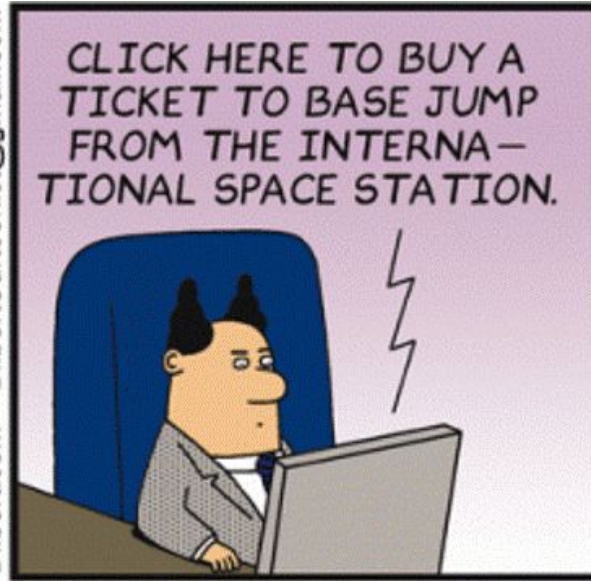
OUTLINE

- What is Machine Learning?
- Why it is worth studying it?

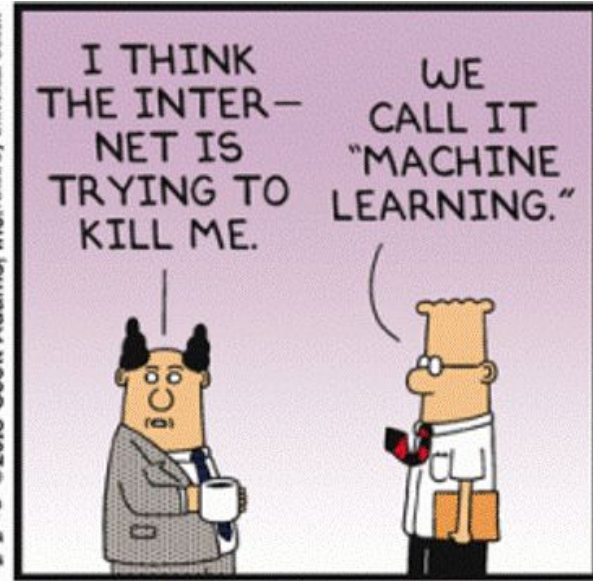
MACHINE LEARNING IS...



Dilbert.com DilbertCartoonist@gmail.com



2-2-13 ©2013 Scott Adams, Inc., /Dist. by Universal Uclick



MACHINE LEARNING IS...

Machine learning is a field of study in artificial intelligence concerned with the development and study of statistical algorithms that can learn from data and generalize to unseen data, and thus perform tasks without explicit instructions.



WIKIPEDIA
The Free Encyclopedia

MACHINE LEARNING IS...

In supervised learning, the algorithm learns from labeled data, where the inputs are paired with corresponding outputs. The goal is to learn a mapping from inputs to outputs, which can then be used to make predictions on new, unseen data.

In unsupervised learning, the algorithm learns from unlabeled data, where there are no predefined output labels. Instead, the algorithm seeks to find patterns, structure, or relationships in the data.

Semi-supervised learning combines elements of both supervised and unsupervised learning, using a small amount of labeled data along with a larger amount of unlabeled data.

Reinforcement learning involves training an algorithm to make sequences of decisions by rewarding it for good decisions and penalizing it for bad ones, based on feedback from its environment.

Machine learning has a wide range of applications, including but not limited to image and speech recognition, natural language processing, recommendation systems, medical diagnosis, and autonomous vehicles.



You

what is machine learning (in a sentence)?



ChatGPT

Machine learning is the field of study that gives computers the ability to learn and improve from experience without being explicitly programmed.



Message ChatGPT...



ChatGPT can make mistakes. Consider checking important information.



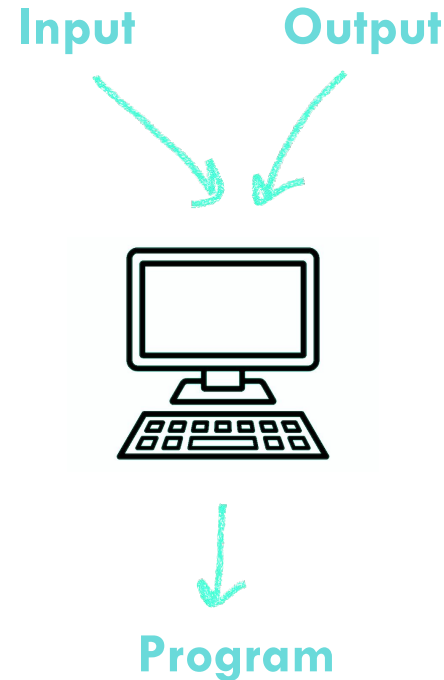
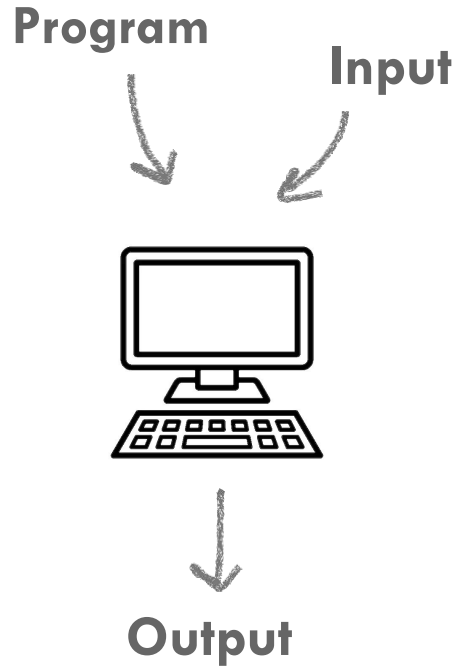
MACHINE LEARNING IS...



Machine learning is the science of getting computers to act without being explicitly programmed.

Arthur Samuel (1959)

PROGRAMMING VS MACHINE LEARNING



PROGRAMMING VS MACHINE LEARNING

Programming:

- You write **explicit rules** and logic
- Input + code $\rightarrow$ output
- Behavior is fully determined by the developer
- Best for well-defined, rule-based tasks

Machine Learning:

- The system **learns patterns from data**
- Input + data $\rightarrow$ model $\rightarrow$ output
- Behavior is learned, not manually coded
- Best for complex tasks (vision, language, prediction)

PROGRAMMING VS MACHINE LEARNING

Programming:

- **Task:** Calculate employee bonuses
- **Approach:** write explicit rules (e.g., if sales > €50k → 10% bonus, else 5%)
- **Pipeline:** input (sales numbers) + code (if/else rules) → output (bonus)



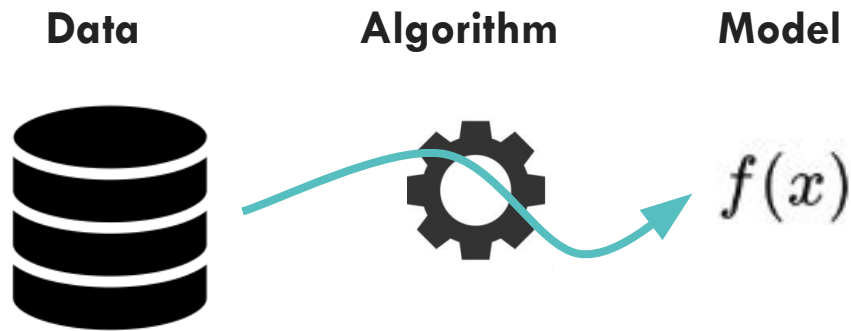
Machine Learning:

- **Task:** Detect spam emails
- **Approach:** train a model on many labeled emails (spam / not spam); model learns patterns (reflecting words, structure, metadata)
- **Pipeline:** input (email) + learned model → output (spam probability)



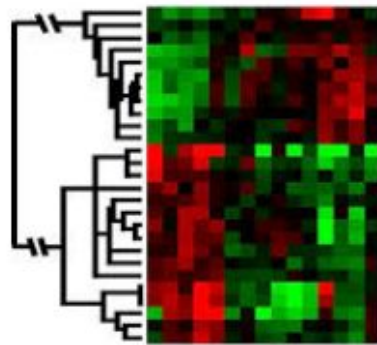
MACHINE LEARNING IDEA

- ML allows computers to acquire knowledge.
- Knowledge is acquired through **algorithms** by learning from **data**.
- Knowledge is represented by a **model**.
- The model is used on future data.



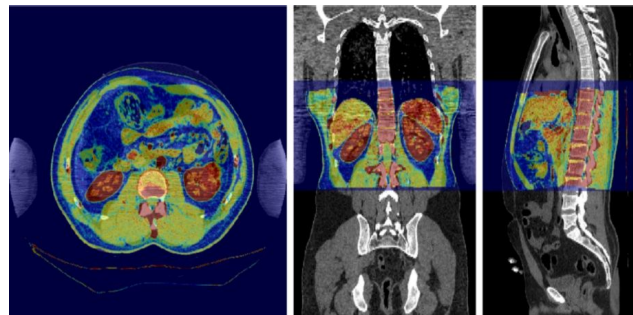
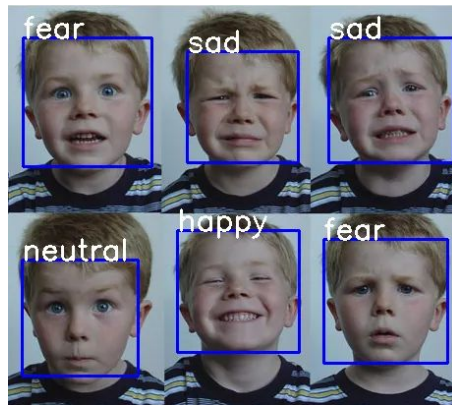
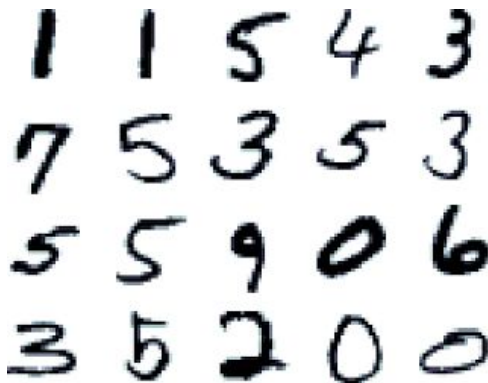
MACHINE LEARNING: WHEN LEARNING FROM DATA

- Learning is not always useful or needed (e.g. calculate employee bonuses)
- ML is used when:
 - Human expertise is unavailable (e.g., navigating on Mars)
 - Humans cannot fully articulate their expertise (e.g., speech recognition)
 - Models need to be tailored to individuals (e.g., personalized medicine)
 - Models rely on massive datasets (e.g., genomics)



MACHINE LEARNING: WHEN LEARNING FROM DATA

- **Recognizing** patterns:
 - Handwritten digits
 - Facial identities or facial expressions
 - Medical images



MACHINE LEARNING: WHEN LEARNING FROM DATA

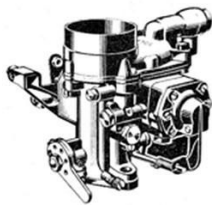
- **Generating** patterns:
 - Generating images or motion sequences



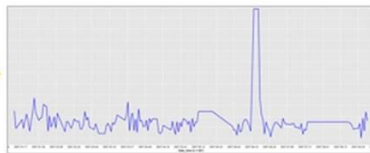
MACHINE LEARNING: WHEN LEARNING FROM DATA

- **Recognizing anomalies**

- e.g., unusual patterns of sensor readings



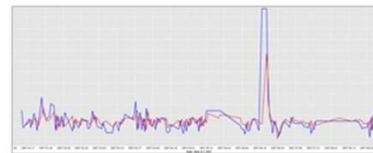
System with
sensors



Time Series from
sensors



Training of
Machine Learning
model

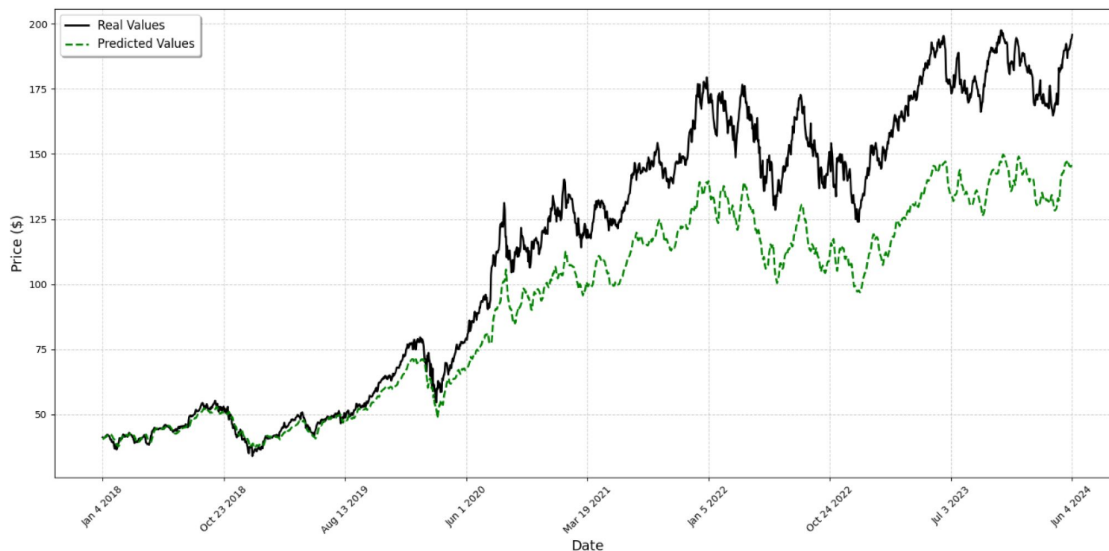


Predicted
Time Series

MACHINE LEARNING: WHEN LEARNING FROM DATA

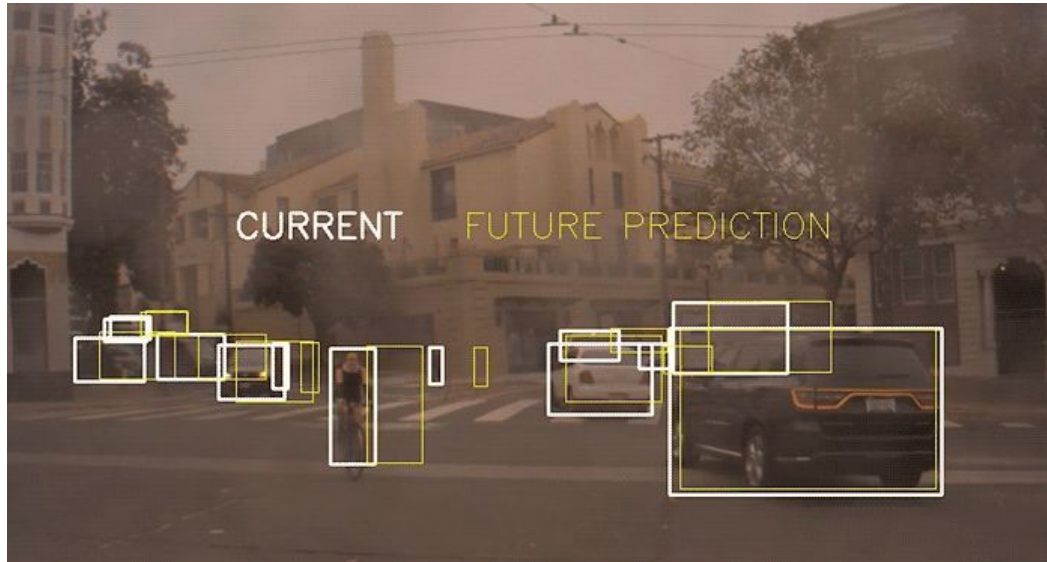
Prediction:

- e.g. future stock prices



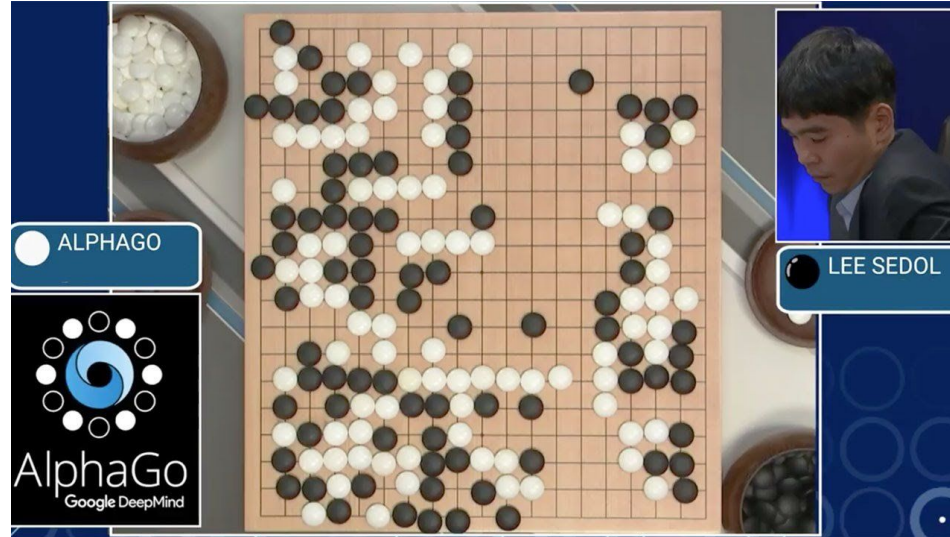
MACHINE LEARNING: WHEN LEARNING FROM DATA

- **Prediction:**
 - Autonomous driving

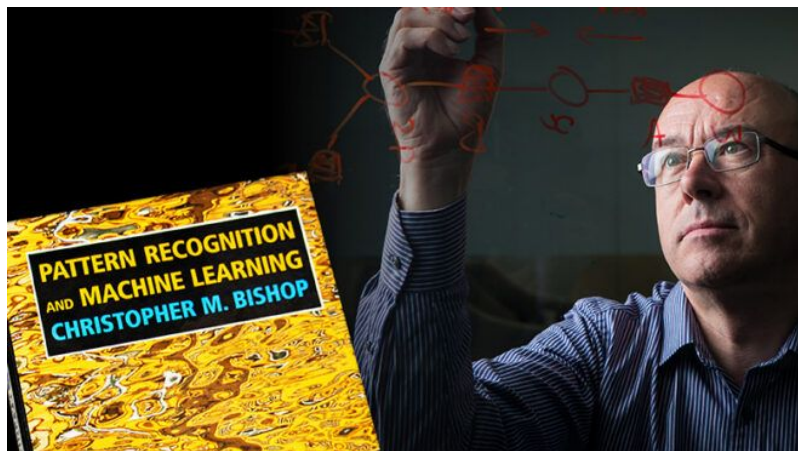


MACHINE LEARNING: WHEN LEARNING FROM DATA

- **Predict best moves**
 - Play Go against humans



MACHINE LEARNING IS...



It is concerned with the **automatic discovery of regularities in data** through the use of computer algorithms and with the use of these regularities to take actions.

Christopher M. Bishop

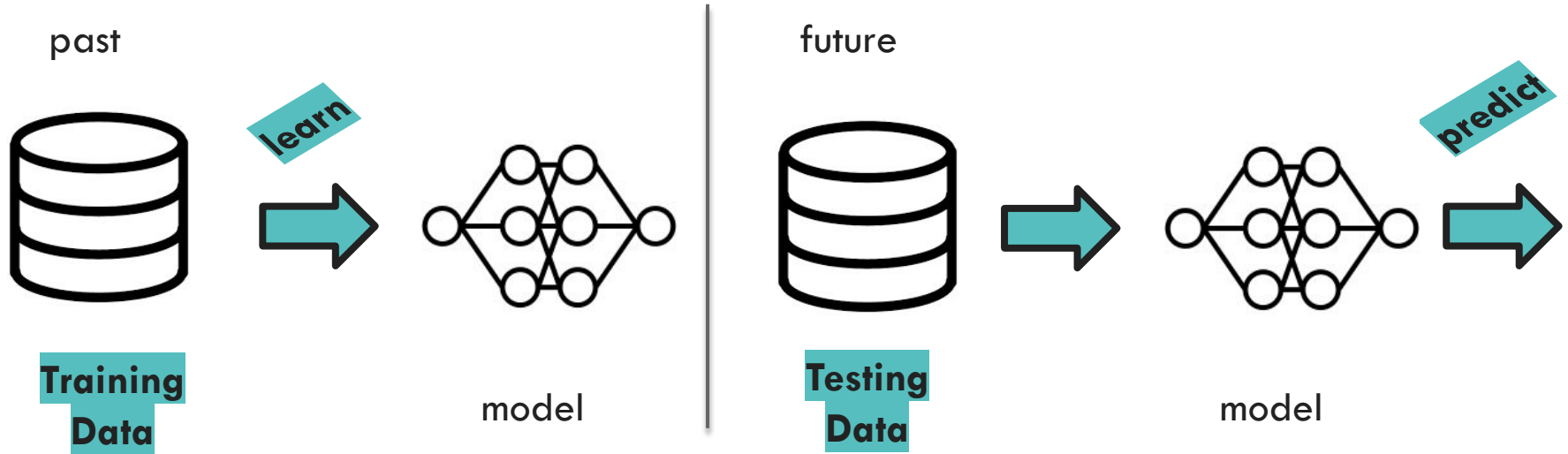
MACHINE LEARNING IS...



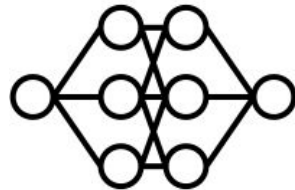
The goal of machine learning is to develop methods that can **automatically detect patterns in data**, and then to use the uncovered patterns to **predict future data** or other outcomes of interest.

Kevin P. Murphy

MACHINE LEARNING IS...



AN EXAMPLE: TRAINING PHASE



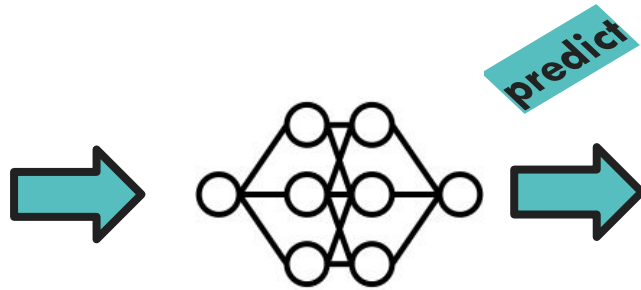
model

**Training
Data**

AN EXAMPLE: TEST PHASE



Testing
Data



model

"grocery store"

MACHINE LEARNING IS...



A computer program is said to learn from **experience E** with respect to some class of **tasks T** and performance **measure P**, if its performance at tasks in T, as measured by P, improves with experience E.”.

Tom Mitchell (1970)

MACHINE LEARNING IS...

- Machine Learning is the study of algorithms that:
 - improve their performance P
 - at some task T
 - with experience E
- A well-defined learning task is given by **a triplet $\langle T, P, E \rangle$** .
- *Can you give some examples?*

MACHINE LEARNING IS...

A well-defined learning task is given by $\langle T, P, E \rangle$.

T: Recognizing handwritten words

P: Percentage of words correctly classified

E: Database of human-labeled images of handwritten words



MACHINE LEARNING IS...

A well-defined learning task is given by $\langle T, P, E \rangle$.

T: Driving on four-lane highways using vision sensors

P: Average distance traveled before a human-judged error

E: A sequence of images and steering commands recorded while observing a human driver.



MACHINE LEARNING IS...

A well-defined learning task is given by $\langle T, P, E \rangle$.

T: Categorize email messages as spam or legitimate.

P: Percentage of email messages correctly classified.

E: Database of emails, some with human-given labels

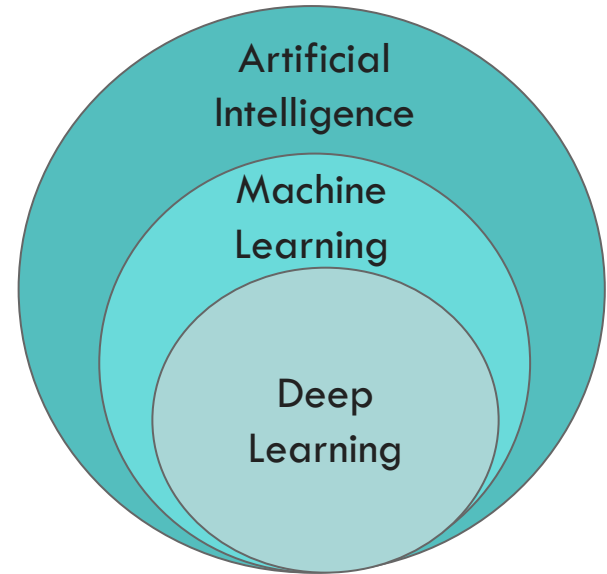


RELATED DISCIPLINES

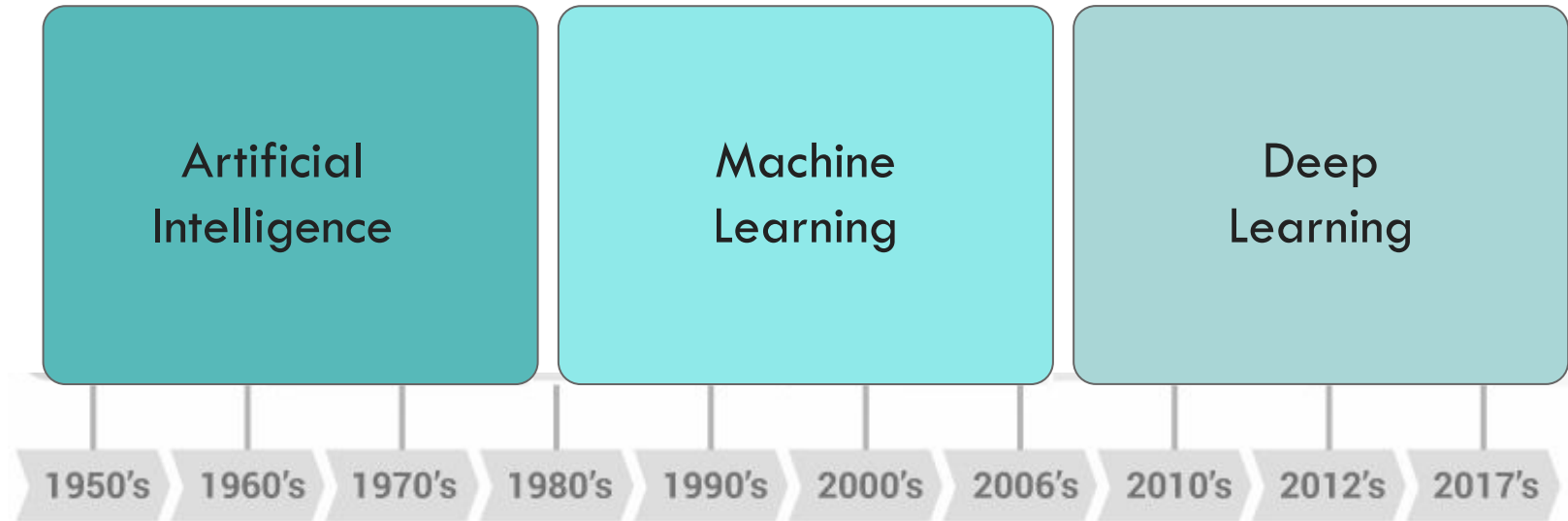
- **data mining**: data analysis, not prediction, though often involves some shared techniques
- **statistical inference**: the process of using data analysis to infer properties of an underlying probability distribution.
- **pattern recognition** in engineering: automatic discovery of regularities in data through the use of computer algorithms and with the use of these regularities to take actions such as classifying the data into different categories
- **signal processing** in electrical engineering
- **optimization**

RELATED DISCIPLINES

- **Artificial Intelligence (AI):** The field of computer science that enables machines to perform tasks that typically require human intelligence, such as reasoning, perception, and decision-making.
- **Machine Learning (ML):** A subset of AI where systems automatically learn patterns and make predictions from data without being explicitly programmed.
- **Deep Learning (DL):** A subset of ML that uses multi-layered neural networks to automatically learn hierarchical representations from large amounts of data.



MACHINE LEARNING, AI & DEEP LEARNING



ARTIFICIAL INTELLIGENCE IS...



Our ultimate objective is to make programs that **learn from their experience** as effectively as humans do.

J. McCarthy, 1958

MACHINE LEARNING IS...



Machine learning is the science of getting computers to act without being explicitly programmed.

Arthur Samuel (1959)

MACHINE LEARNING SUCCESS STORIES

Face detection (2002)



MACHINE LEARNING SUCCESS STORIES

Pedestrian Detection (2005)



MACHINE LEARNING SUCCESS STORIES

Body Tracking in RGB-D (2012)



DEEP LEARNING IS...

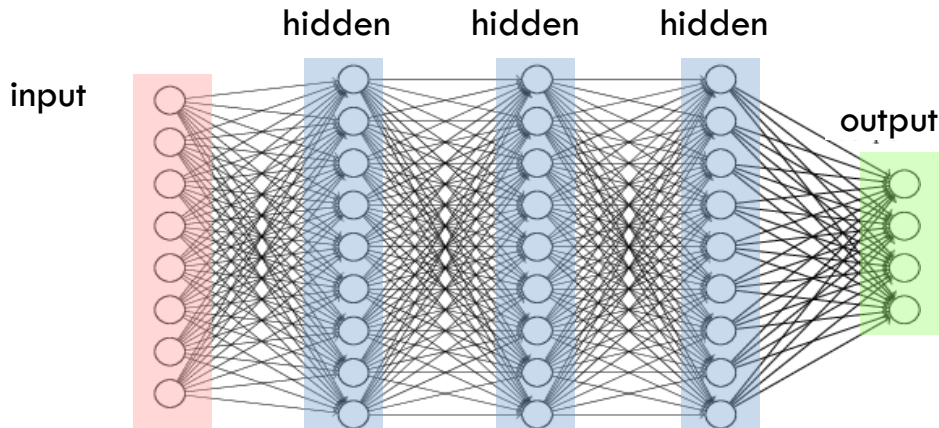


“Deep learning allows computational models that are composed of **multiple processing layers** to **learn representations of data** with multiple levels of abstraction”

Nature

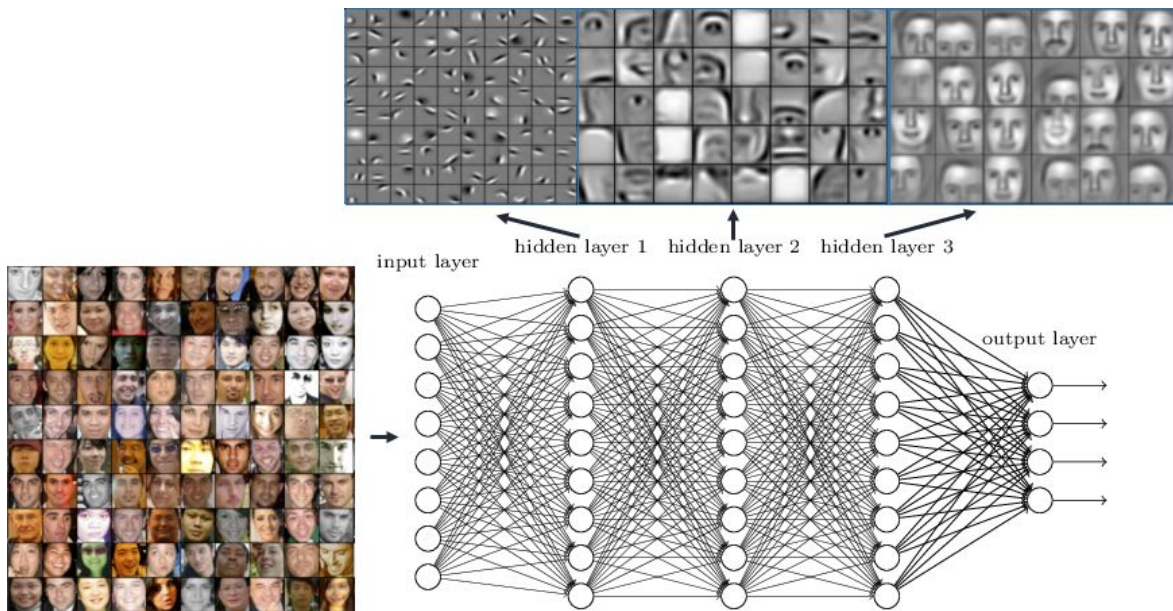
DEEP LEARNING IS...

Deep Learning means using a neural network with **several layers** of nodes between input and output



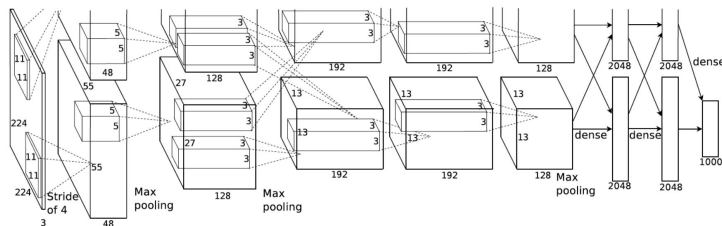
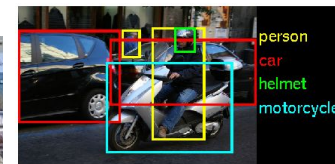
DEEP LEARNING IS...

The series of layers between input and output compute **relevant features** automatically in a series of stages, just as our brains seem to.

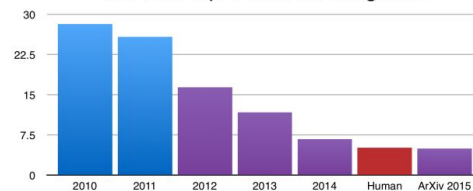


DEEP LEARNING SUCCESS STORIES

Object recognition (2012)

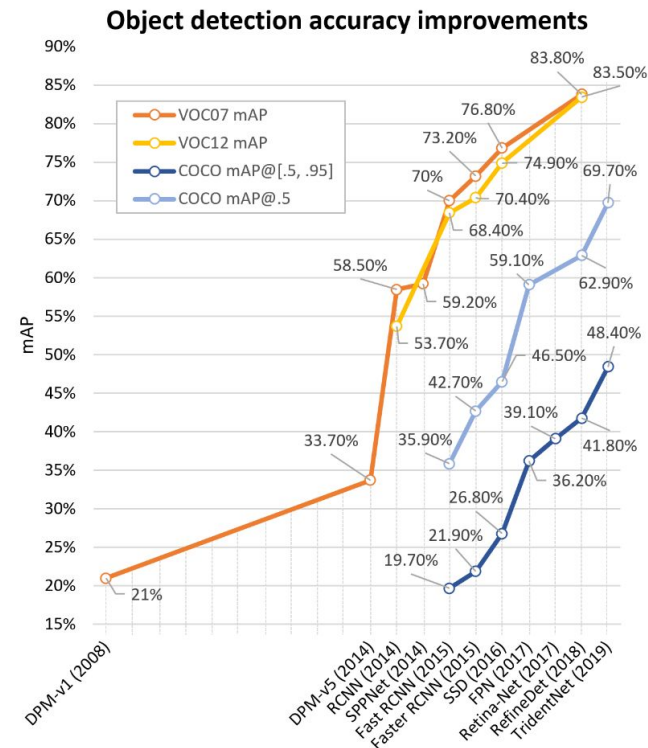
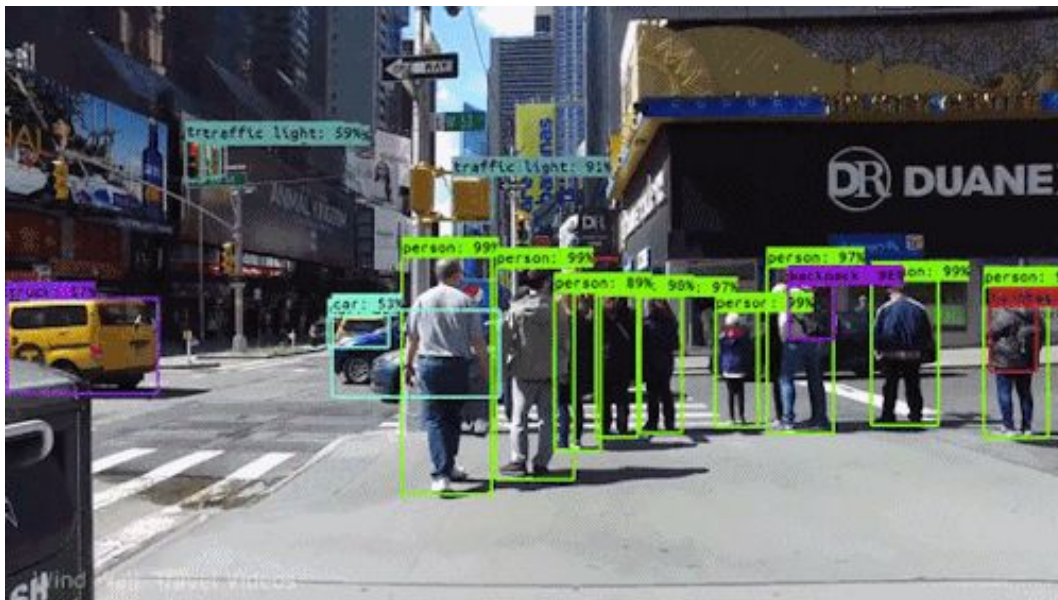


ILSVRC top-5 error on ImageNet



DEEP LEARNING SUCCESS STORIES

Object detection

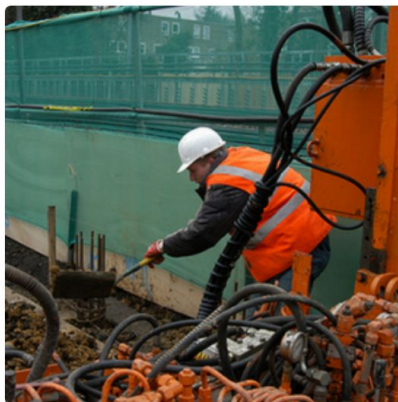


DEEP LEARNING SUCCESS STORIES

Image captioning (2015)



"man in black shirt is playing guitar."



"construction worker in orange safety vest is working on road."



"two young girls are playing with lego toy."

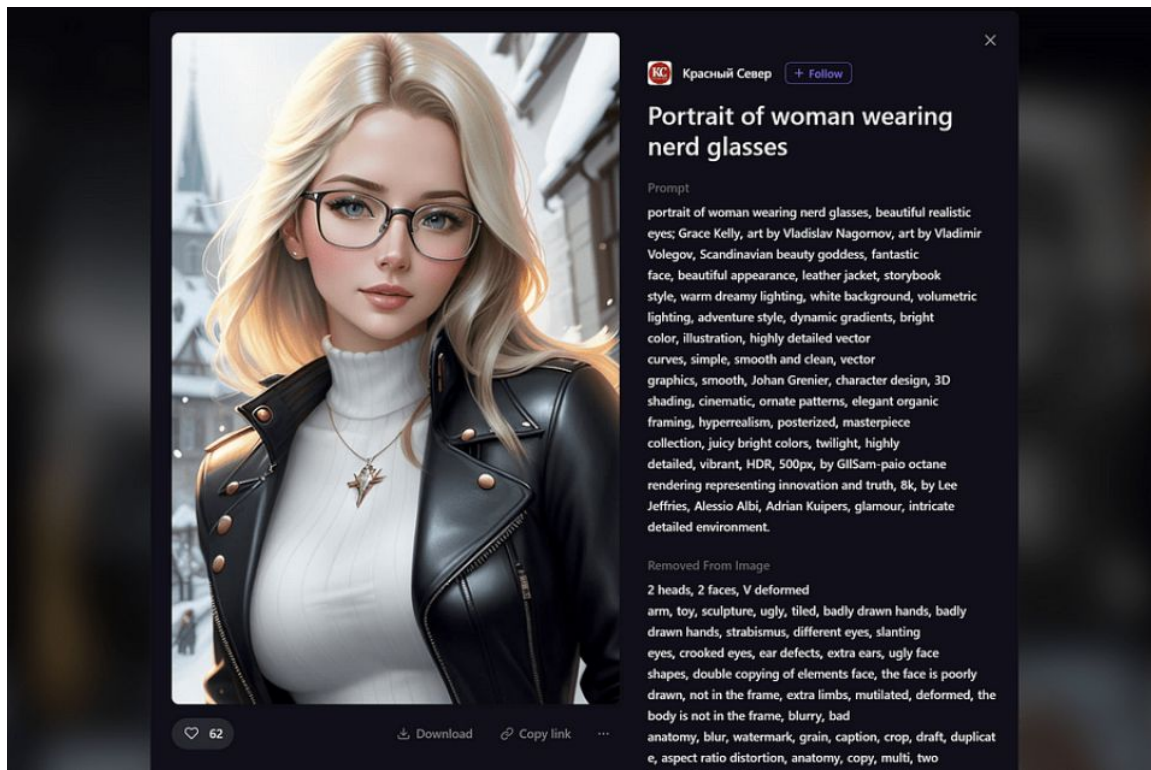
DEEP LEARNING SUCCESS STORIES

Image synthesis (2016)



DEEP LEARNING SUCCESS STORIES

Text to Image synthesis (2022)



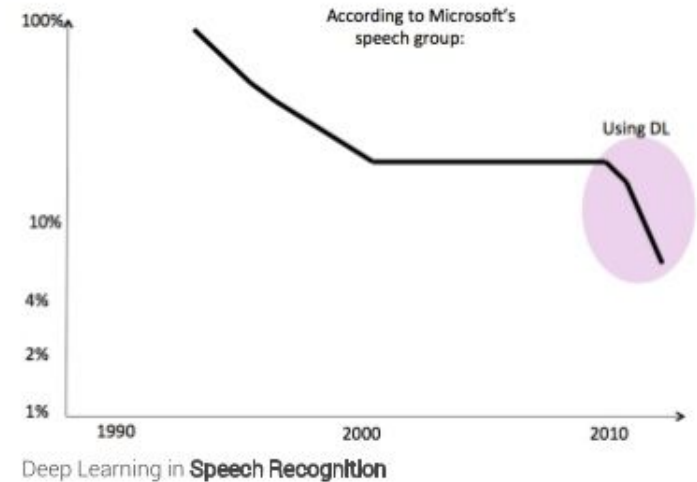
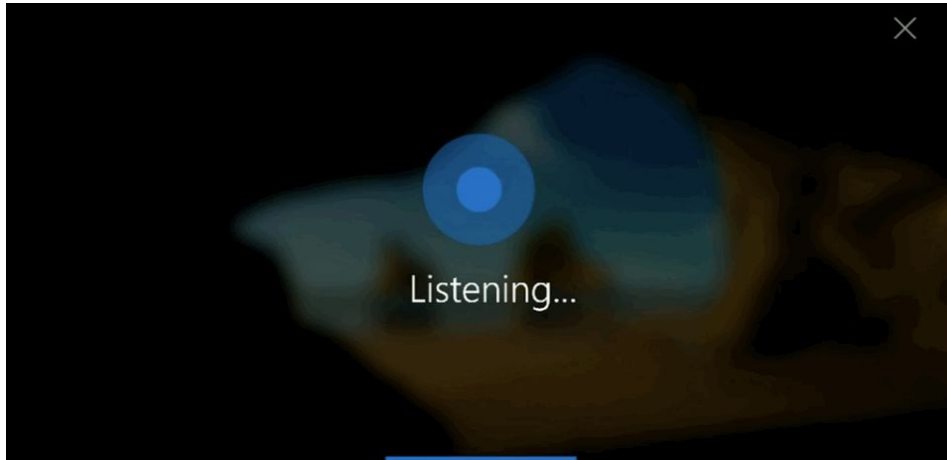
DEEP LEARNING SUCCESS STORIES

Video synthesis
(2024)



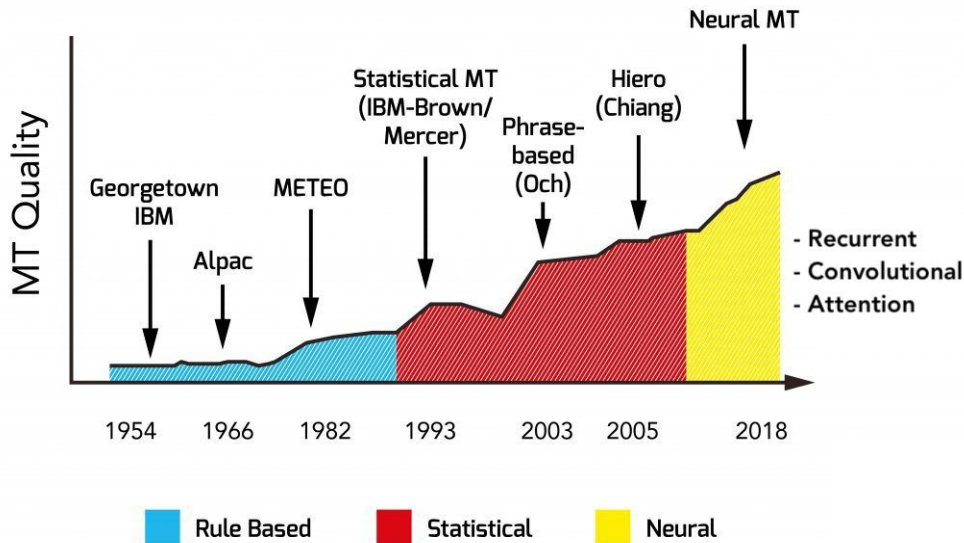
DEEP LEARNING SUCCESS STORIES

Speech Recognition (2009)



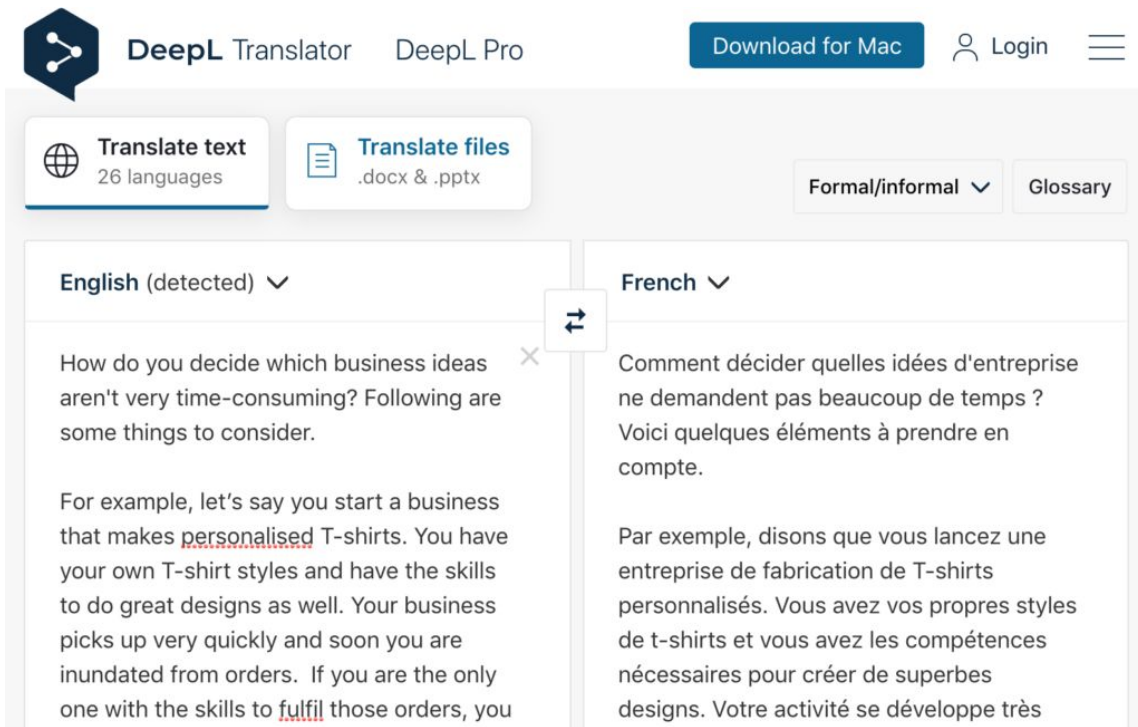
DEEP LEARNING SUCCESS STORIES

Neural Machine Translation (2014)



DEEP LEARNING SUCCESS STORIES

Neural Machine Translation



The screenshot displays the DeepL Translator web application. At the top, the navigation bar includes the DeepL logo, the text 'DeepL Translator' and 'DeepL Pro', a 'Download for Mac' button, a 'Login' button with a user icon, and a hamburger menu icon. Below the navigation bar, there are two main buttons: 'Translate text' (with a globe icon and '26 languages') and 'Translate files' (with a document icon and '.docx & .pptx'). To the right of these buttons are two dropdown menus: 'Formal/informal' and 'Glossary'. The main content area is split into two columns. The left column is labeled 'English (detected)' and contains two paragraphs of text. The right column is labeled 'French' and shows the translated version of the text. A double-headed arrow icon is positioned between the two columns, indicating the translation direction. The text in the English column is: 'How do you decide which business ideas aren't very time-consuming? Following are some things to consider. For example, let's say you start a business that makes personalised T-shirts. You have your own T-shirt styles and have the skills to do great designs as well. Your business picks up very quickly and soon you are inundated from orders. If you are the only one with the skills to fulfil those orders, you'. The text in the French column is: 'Comment décider quelles idées d'entreprise ne demandent pas beaucoup de temps ? Voici quelques éléments à prendre en compte. Par exemple, disons que vous lancez une entreprise de fabrication de T-shirts personnalisés. Vous avez vos propres styles de t-shirts et vous avez les compétences nécessaires pour créer de superbes designs. Votre activité se développe très'.

DeepL Translator DeepL Pro

Download for Mac Login

Translate text 26 languages

Translate files .docx & .pptx

Formal/informal Glossary

English (detected)

French

How do you decide which business ideas aren't very time-consuming? Following are some things to consider.

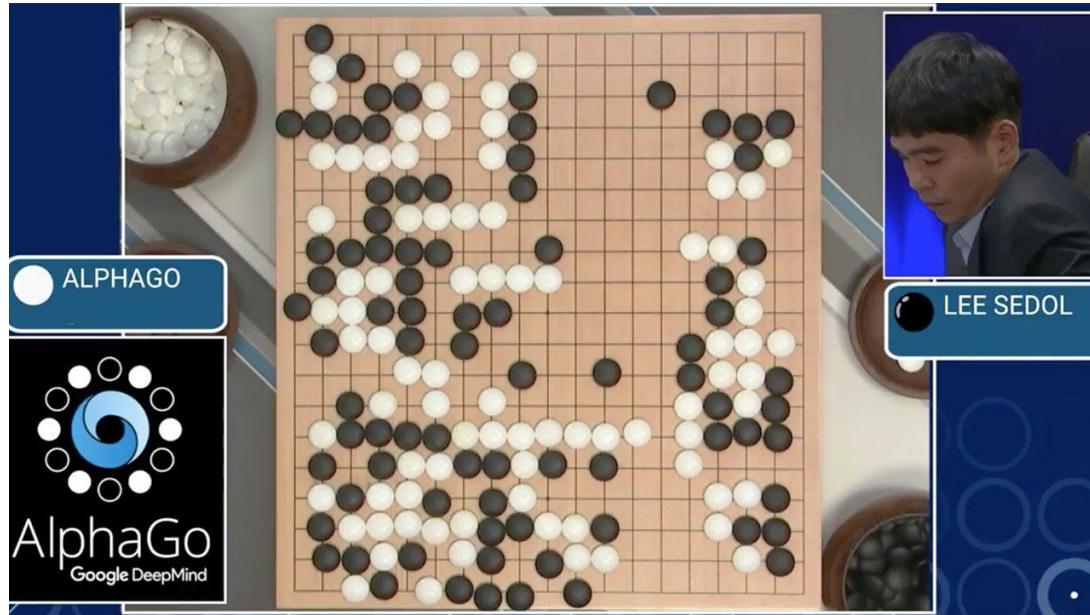
For example, let's say you start a business that makes personalised T-shirts. You have your own T-shirt styles and have the skills to do great designs as well. Your business picks up very quickly and soon you are inundated from orders. If you are the only one with the skills to fulfil those orders, you

Comment décider quelles idées d'entreprise ne demandent pas beaucoup de temps ? Voici quelques éléments à prendre en compte.

Par exemple, disons que vous lancez une entreprise de fabrication de T-shirts personnalisés. Vous avez vos propres styles de t-shirts et vous avez les compétences nécessaires pour créer de superbes designs. Votre activité se développe très

DEEP LEARNING SUCCESS STORIES

AlphaGO (2015)



DEEP LEARNING SUCCESS STORIES

ChatGPT (2022)



What is ChatGPT?



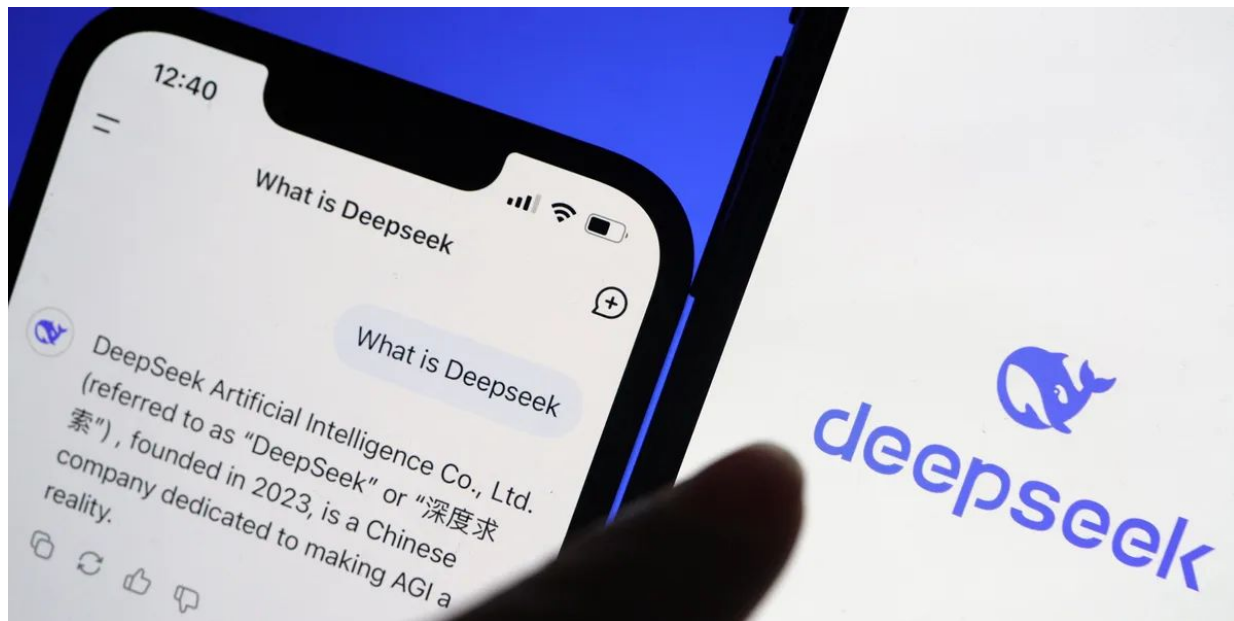
ChatGPT is a chatbot designed to assist with customer support, conversation, and information gathering tasks. It uses natural language processing and GPT-3 technology to generate responses that are tailored to the user's input. ChatGPT can handle a wide range of queries and provide answers in real-time, making it a valuable tool for businesses looking to improve their customer support experience.

< 2 / 2 >



DEEP LEARNING SUCCESS STORIES

Deepseek (2025)



DEEP LEARNING REVOLUTION: WHY NOW?

- Flood of available **data**
- Increased **computational power**
- Growing number of **ML algorithms and theory** developed by researchers
- Increased **support from industry**

DEEP LEARNING REVOLUTION: BENEFITS & SECTORS

- **Healthcare:** Faster diagnosis, drug discovery
 - **Safety & Security:** Disaster prediction, public safety monitoring
 - **Efficiency & Productivity:** Automation of repetitive tasks, optimized logistics, energy management
 - **Education:** Personalized learning, intelligent tutoring systems
 - **Accessibility:** Assistive technologies for differently-abled individuals
 - **Environment:** Climate modeling, smart agriculture
- and many more...

DEEP LEARNING REVOLUTION: RISKS

- **Bias:** AI can perpetuate biases in decision-making, impacting fairness.
- **Privacy issues:** AI technologies raise privacy issues through the collection and analysis of massive amounts of personal data.
- **Job displacement:** Automation thanks to ML may lead to widespread job loss, particularly in industries relying on routine tasks.



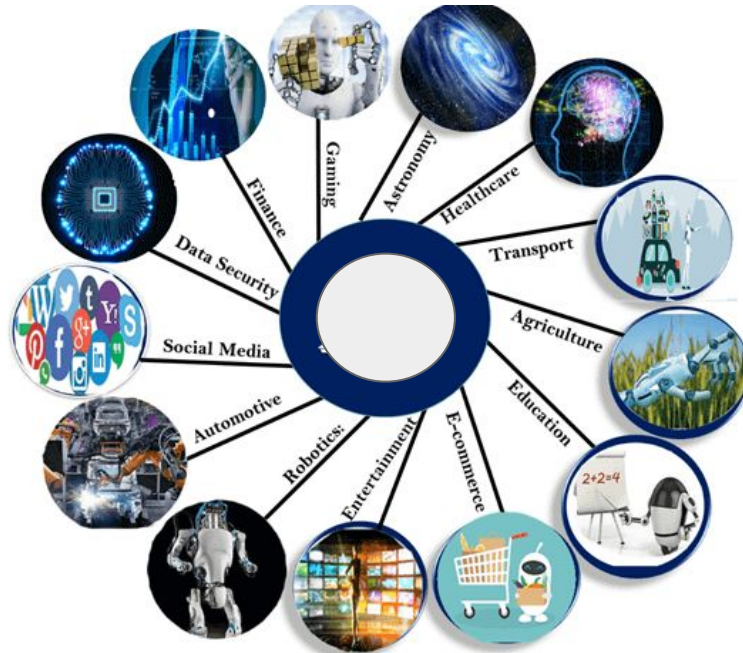
WHY STUDY MACHINE LEARNING?

Machine Learning is everywhere



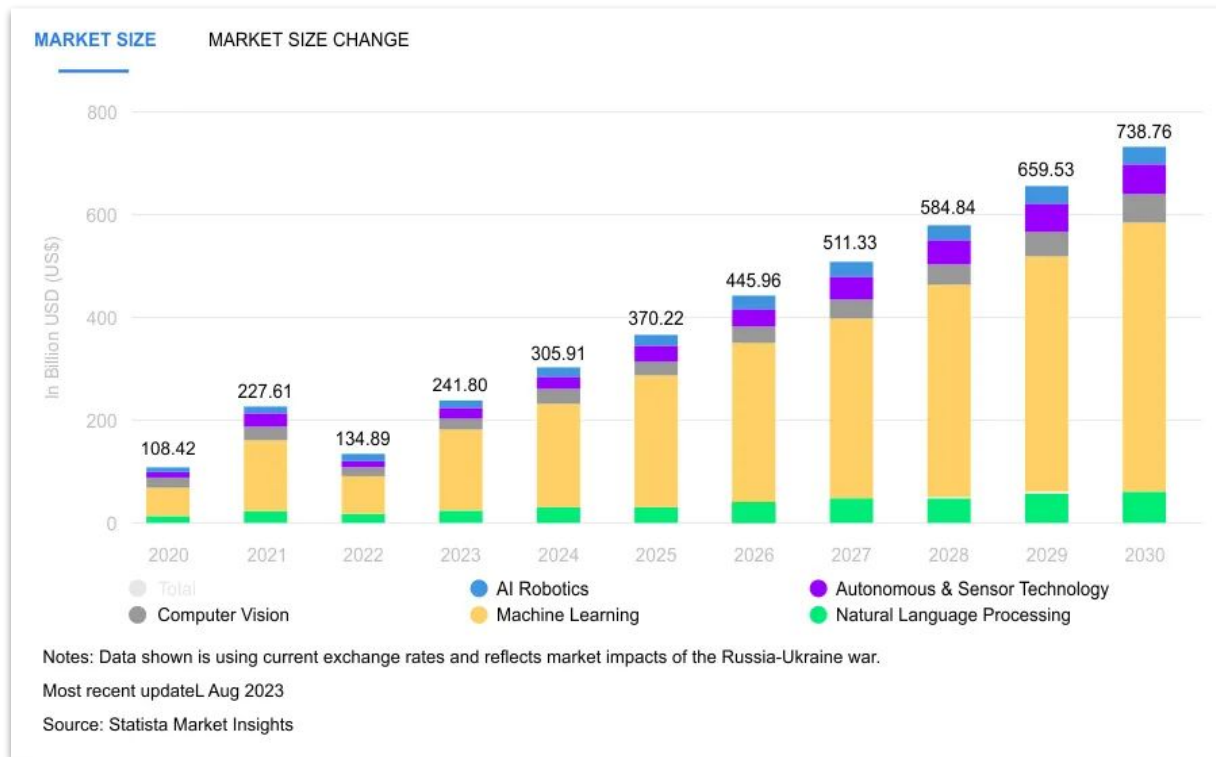
WHY STUDY MACHINE LEARNING?

Machine Learning is everywhere



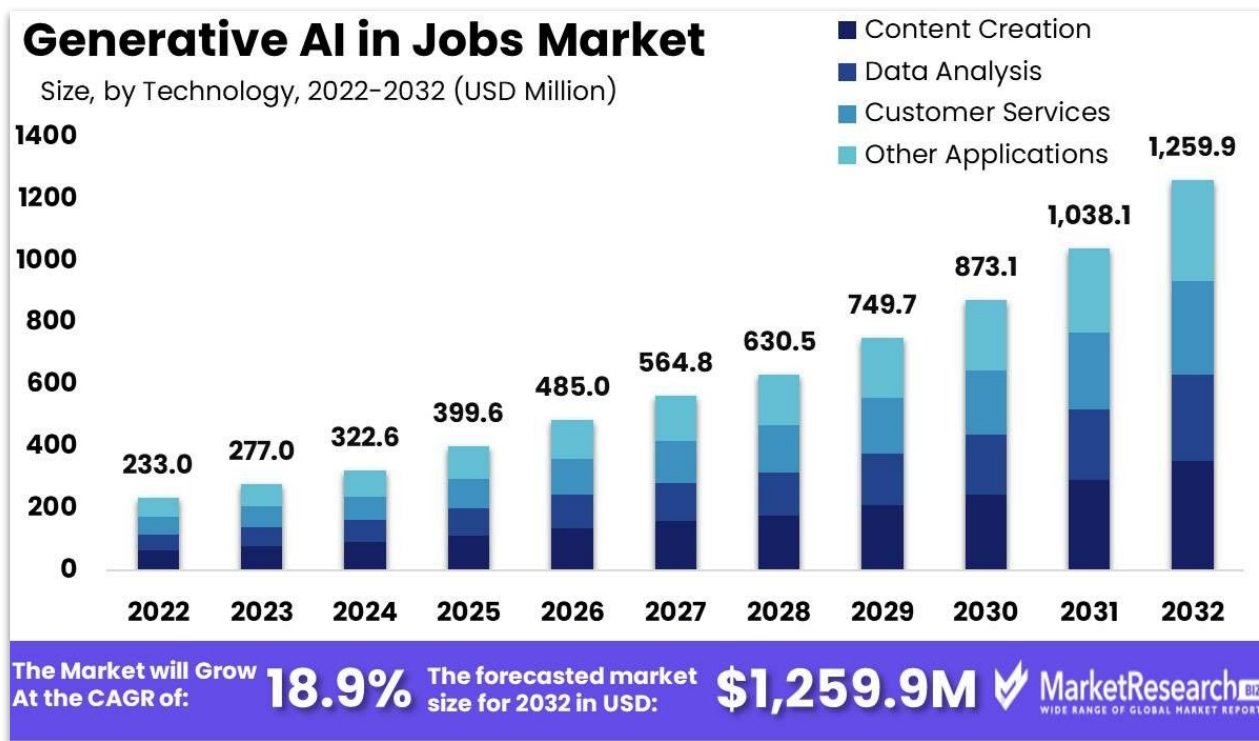
WHY STUDY MACHINE LEARNING?

Jobs



WHY STUDY MACHINE LEARNING?

Jobs



WHY STUDY MACHINE LEARNING?

Jobs

MENU

Agenda Digitale

Cittadinanza digitale

Sicurezza Informatica

Sanità digitale

Industry 4.0/Innovazione in azienda

Infrastrutture digitali

Procurement dell'innovazione

Fatturazione elettronica

Documenti digitali

ISCRIVITI SUBITO

I DATI

IA e lavoro: chi rischia di più in Italia e quali competenze servono

Home > Cultura E Società Digitali

f in X e

Partecipa al dibattito

L'impatto dell'IA sul lavoro non si limita a “togliere” o “creare” posti: cambia soprattutto come si lavora. La differenza rispetto all'automazione tradizionale è che l'IA entra nelle attività cognitive, spesso ad alta qualifica, come alleato operativo. Il nodo centrale diventa la trasformazione delle competenze, tecniche e umane

Pubblicato il 12 feb 2026

Rossella Fasola
Public Affairs Director Randstad

Lorenzo Maternini
Direttore Fondazione Randstad AI & Humanities

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QUESTIONS?



SUGGESTED RESOURCES

ARTIFICIAL INTELLIGENCE AND LIFE IN 2030

ONE HUNDRED YEAR STUDY ON ARTIFICIAL INTELLIGENCE | REPORT OF THE 2015 STUDY PANEL | SEPTEMBER 2016

PREFACE

The One Hundred Year Study on Artificial Intelligence, launched in the fall of 2014, is a long-term investigation of the field of Artificial Intelligence (AI) and its influences on people, their communities, and society. It considers the science, engineering, and deployment of AI-enabled computing systems. As its core activity, the Standing Committee that oversees the One Hundred Year Study forms a Study Panel every five years to assess the current state of AI. The Study Panel reviews AI's progress in the years following the immediately prior report, envisions the potential advances that lie ahead, and describes the technical and societal challenges and opportunities these advances raise, including in such arenas as ethics, economics, and the design of systems compatible with human cognition. The overarching purpose of the One Hundred Year Study's periodic expert review is to provide a collected and connected set of reflections about AI and its influences as the field advances. The studies are expected to develop syntheses and assessments that provide expert-informed guidance for directions in AI research, development, and systems design, as well as programs and policies to help ensure that these systems broadly benefit individuals and society.¹

The One Hundred Year Study is modeled on an earlier effort informally known as the "AAAI Asilomar Study." During 2008-2009, the then president of the Association for the Advancement of Artificial Intelligence (AAAI), Eric Horvitz, assembled a group of AI experts from multiple institutions and areas of the field, along with



The overarching purpose of the One Hundred Year Study's periodic expert review is to provide a collected and connected set of reflections about AI and its influences as the field advances.

[Link](#)

SUGGESTED RESOURCES

The 2025 AI Index Report

[Read the 2025 AI Index Report](#)[Watch Seminar](#)

AI's influence on society has never been more pronounced.

At Stanford HAI, we believe AI is poised to be the most transformative technology of the 21st century. But its benefits won't be evenly distributed unless we guide its development thoughtfully. The AI Index offers one of the most comprehensive, data-driven views of artificial intelligence. Recognized as a trusted resource by global media, governments, and leading companies, the AI Index equips policymakers, business leaders, and the public with rigorous, objective insights into AI's technical progress, economic influence, and societal impact.

New this Year: The Official Chinese Version of the 2025 AI Index Report

[Read the translation](#)